

PRISMA 2020 Checklist

Manuscript: “A Systematic Review of Deep Knowledge Tracing (2015–2025): Toward Responsible AI for Education”

Section/topic	#	Checklist item	Location	Notes
TITLE				
Title	1	Identify the report as a systematic review.	Title	“A Systematic Review of Deep Knowledge Tracing...”
ABSTRACT				
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract	Structured abstract with objectives, methods, results, conclusions
INTRODUCTION				
Rationale	3	Describe the rationale in the context of existing knowledge.	Section 1	Existing reviews lack RAI perspective (Section 1.3)
Objectives	4	Provide an explicit statement of the objective(s) or question(s).	Section 1.4	Four research questions (RQ1–RQ4)
METHODS				
Eligibility criteria	5	Specify the inclusion and exclusion criteria.	Section 2.2, Table 1	I1–I4, E1–E6 listed
Information sources	6	Specify all databases, registers, and other sources searched.	Section 2.1	Web of Science, Scopus, ScienceDirect, ACM DL, IEEE Xplore, Google Scholar
Search strategy	7	Present the full search strategies for all databases.	Section 2.1; Supp. Mat. A	Core terms in text; full strings in supplement
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria.	Section 2.3	Two-phase screening; 17% double-coded; discrepancies resolved by discussion
Data collection process	9	Specify the methods used to collect data from reports.	Section 2.5	Standardized extraction form; two researchers
Data items	10a	List and define all outcomes for which data were sought.	Section 2.5	Publication details, technical specs, rigor indicators, outcomes
	10b	List and define all other variables for which data were sought.	Section 2.5; Appendix Table	Model category, datasets, metrics, data quality, stability, interpretability
Study risk of bias	11	Specify the methods used for assessing risk of bias.	Section 2.4	Five custom quality appraisal criteria; scoring 0/0.5/1
Effect measures	12	Specify for each outcome the effect measure(s) used.	N/A	Qualitative synthesis; no meta-analysis
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis.	Section 2.5	Deductive (RQ1) and inductive thematic analysis (RQ2–4)
	13b	Describe any methods required to prepare the data for synthesis.	Section 2.5	Taxonomy adapted from Abdelrahman et al.
	13c	Describe any methods used to tabulate or visually display results.	Sections 3.1–3.4	Tables, parallel sets diagram, heatmap

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Reporting bias assessment Certainty assessment	13d	Describe any methods used to synthesize results.	Section 2.5	Content analysis + thematic analysis
	13e	Describe any methods used to explore possible causes of heterogeneity.	Section 4.5	Cross-cutting patterns across RAI dimensions
	13f	Describe any sensitivity analyses conducted.	Section 2.4; Supplementary Material B	Sensitivity analysis on quality appraisal thresholds
	14	Describe any methods used to assess risk of reporting biases.	Section 4.6 (Limitations)	English-only, keyword scope limitations acknowledged
	15	Describe any methods used to assess certainty in the body of evidence.	N/A	Not applicable (qualitative synthesis)
RESULTS				
Study selection	16a	Describe the results of the search and selection process.	Section 2.3, Fig. 2	PRISMA flow diagram (958+89 identified, 83 included)
	16b	Cite studies that appeared to meet criteria but were excluded, and explain why.	Section 2.3	Exclusion counts per criterion reported
Study characteristics	17	Cite each included study and present its characteristics.	Appendix Table	83 studies with key dimensions (6-part table)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Material B	Per-study QA1–QA5 scores for all 83 studies
Results of individual studies	19	Present results of all individual studies.	Appendix Table	Summary table with per-study characteristics
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias.	Sections 3.1–3.4	Results organized by RQ
	20b	Present results of all statistical syntheses conducted.	Sections 3.1–3.4	Counts, percentages, cross-tabulations
	20c	Present results of all investigations of possible causes of heterogeneity.	Section 4.5	Cross-cutting patterns by model category
	20d	Present results of all sensitivity analyses conducted.	Section 2.4; Supplementary Material B	Removing medium-quality studies does not change key findings
Reporting biases	21	Present assessments of risk of reporting biases.	Section 4.6	Limitations section
Certainty of evidence	22	Present assessments of certainty in the body of evidence.	N/A	Not applicable
DISCUSSION				
Discussion	23a	Provide a general interpretation of results in context of other evidence.	Sections 4.1–4.5	Four subsections + cross-cutting patterns
	23b	Discuss any limitations of the evidence included in the review.	Section 4.6	English-only, taxonomy subjectivity, keyword scope
	23c	Discuss any limitations of the review processes used.	Section 4.6	No meta-analysis, subjective classification
	23d	Discuss implications of the results for practice, policy, and future research.	Sections 4.1–4.5; Section 5	Throughout discussion and conclusion
OTHER INFORMATION				

Section/topic	#	Checklist item	Location	Notes
Registration and protocol	24a	Provide registration information, including register name and number.	N/A	Not registered
	24b	Indicate where the review protocol can be accessed.	N/A	Not registered
	24c	Describe and explain any amendments to the protocol.	N/A	–
Support	25	Describe sources of financial or non-financial support.	Acknowledgements	Estonian Research Council (PRG2215)
Competing interests	26	Declare any competing interests of review authors.	Declarations	No competing interests
Availability of data, code, and other materials	27	Report which of the following are publicly available and where: template data collection forms; data extracted from included studies; data used for analyses; analytic code; any other materials.	Data/Code availability	https://github.com/kiote/dkt-systematic-review-data

From: Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi:10.1136/bmj.n71.

All checklist items have been addressed.